A separator after the six-point head at odd degree

Human Compression project: L044 continuation

Let $G \leq \mathrm{Alt}(\Omega)$ be transitive with connected string C-marking $\rho_0, \ldots, \rho_{r-1}$, all maximal omissions intransitive, $r \geq 6$ and $|\Omega| = 2r + 1$. Its first three marks are supported and transitive on $D = \{1, \ldots, 6\}$. Assume the actual interfaces

$$\rho_3 = (6\ \beta)\delta, \qquad G_{\geq 3} = \mathrm{Alt}(\Omega \setminus \{1, \ldots, 5\}), \qquad \rho_4|_D = 1, \qquad G_{\geq 5}|_{D \cup \{\beta\}} = 1.$$

Here δ is an odd involution supported outside $D \cup \{\beta\}$. For every $j \geq 5$ assume there are two j-edges joining distinct orbits of the full ambient omission G_j. They may lie in different orbits of $H = G_{\geq 5}$. Put $K = G_{>5}$.

Theorem 1. *There is an index s with*

$$5 \leq s < r - 1, \qquad |\operatorname{Fix}(G_{>s})| \geq 2s + 1.$$

Proof. Use the complete strong simultaneous-witness theorem. It supplies a colour $j \geq 5$ with at most one ambient-eligible edge in each actual H-orbit, and at least two distinct witness orbits. Every witness on which K is intransitive has first active label 5, exactly one edge eligible for its full local j-omission, and is fixed pointwise by δ.

If any witness is K-transitive, the complete transitive-witness separator applies, with its actual whole-orbit transport. Its last active K-label $x \geq 6$ is nonterminal and gives the required bound with $s = x$.

Otherwise select two distinct witnesses, and let their last active labels be $x \leq y$. Both contain the colour j, so $x \geq j$. If $x = y$, the complete simultaneous-witness theorem gives $x < r - 1$ and at least $2x + 1$ fixed points; take $s = x$. If $x < y$ and $x \geq j + 2$, apply the complete unequal-witness separator with $s = x$. If $x = j + 1 < y$, use the complete one-step separator with the same $s = x$.

The only remaining case is $x = j < y$. The complete crossing-endpoint separator gives $s = j+1$ when $j \geq 6$, and $s \in \{6, y\}$ when $j = 5$. In each case it explicitly proves $s < r - 1$ and the stated fixed-point bound. All its hypotheses have been supplied by the simultaneous witness theorem. These cases exhaust every possible pair of endpoint labels and every possible first-omission action. □

Attribution and scope. The actual head and orbit-deficit strategy originate in the proof of Proposition 5.31 of Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1. This assembly uses the complete retained proofs of the five stated separator results and their supporting orbit and parity arguments. It adds only the exhaustive case routing; no new induction, computation or formal statement is required. The fixed points are those of the actual group in the displayed theorem. Applying it to a proper prefix requires a separate proof that the required fixed points survive any later attachment. No complete L044 recovery or whole-original credit is claimed.